

Thermal Energy Transfers

Memorize what temperature, heat and internal energy mean — then recognize which equation or transfer mechanism applies.

CORE CONCEPTS

States of matter

- Solid: fixed shape + volume; particles vibrate about fixed positions.
- Liquid: fixed volume; particles slide past one another.
- Gas: no fixed shape/volume; particles move freely.

Temperature

- Measures the average kinetic energy of particles.
- Hotter → larger average KE → faster particles.
- Use Kelvin in physics equations: $T(K) = T(^{\circ}C) + 273$.

MUST-KNOW FORMULAS

$$\langle E_k \rangle = \frac{3}{2} k_B T \cdot U = \Sigma E_k + \Sigma E_p$$

$$Q = mc\Delta T \cdot Q = mL$$

$$Q = \frac{kA\Delta T}{d} \cdot \lambda_{\max} T = b \cdot I = \frac{P}{(4\pi r^2)} \cdot P = \epsilon \sigma A T^4$$

Constants: $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$ · $b = 2.898 \times 10^{-3} \text{ m} \cdot \text{K}$ · $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

SPECIFIC HEAT vs LATENT HEAT

$Q = mc\Delta T$

- Use when the substance is changing temperature but staying in one phase.
- c = energy needed per kg per K.

$Q = mL$

- Use during a phase change.
- Temperature stays constant because energy changes particle separation/bonds.
- Fusion = melting/freezing; vaporization = boiling/condensing.

TRANSFER MECHANISMS

Conduction · Convection · Radiation

- Conduction: energy transfer through matter without bulk movement; strongest in solids/metals.
- Convection: bulk movement of a fluid; warmer, less-dense fluid rises while cooler, denser fluid sinks.
- Radiation: electromagnetic waves; needs no medium, so it crosses a vacuum.
- Heat naturally moves hot → cold.

BLACK-BODY MEMORY

Black body

- Perfect absorber and maximum emitter; $\epsilon = 1$.
- As T rises: total emitted power rises and the peak shifts to shorter λ .

Exam traps

- Doubling T does not double speed: $v \propto \sqrt{T}$.
- T^4 matters: doubling absolute $T \rightarrow 16\times$ radiated power.
- Use Kelvin for Wien, Stefan-Boltzmann and gas/thermal equations.

STUDY METHOD: cover the right side / formula boxes → recall from memory → check → repeat.